

- Microwaves are electromagnetic waves whose freq. range from 1 GHz to 1000 GHz.
- MW's are so called since they are defined in terms of their wavelength in micro ranges (very short)
- Microwave Bands are given.

→ ADVANTAGES:-

1) 1ed BW availability : → Due to micro ranges of wavelength.

and the advantage of large bandwidth is that the freq. range of information channel will be a small percentage of carrier freq and more information can be transmitted in mw freq. ranges. Microwave freq ranges are very useful as the small freq. ranges are very crowded.

2) Improved Directive properties : →

As we know, that freq is proportional to directivity and inversely proportional to decreases, Beamwidth.

$$\therefore \theta \text{ (Beam Width) of Radiation} \propto \frac{\lambda \text{ (wavelength)}}{D \text{ (Directivity)}}$$

so at mw freq., antenna size (Diameter) of several wavelength lead to smaller beam widths and an extremely directed beam, like optical lens focusses light rays.

$\therefore$ Gain (Power) $\propto 1/\lambda^2$; so at mw freq high gain antenna can be designed.

3) fading effect and reliability : →

fading is due to variation in the transmission medium is more effective at low frequency. Due to LOS propagation and high freq's is less fading effect.



4) Power Requirement : $\rightarrow$ T_x / R_x power requirements are pretty low at MW freqs

5) Transparency Property : $\rightarrow$

At MW freqs ϵ_g s are capable of freely propagating through the ionized layers surrounding the earth as well as through the atmosphere.

Wave Guides

A hollow metallic tube of uniform cross section for transmitting electromagnetic wave by successive reflections from the inner walls of the tube is called W.G.

Comparison of W.G. with Transmission line : $\rightarrow$

- | | |
|--|---|
| <ul style="list-style-type: none"> Wave travelling in a W.G. has a phase velocity. At the end of the W.G., wave is reflected | <ul style="list-style-type: none"> 1. In TL, wave attenuated while travelling. 2. |
|--|---|

Types of Wave Guide : $\rightarrow$

MICROWAVE BANDS

L $\rightarrow$ 1 to 2 GHz
 S $\rightarrow$ 2 to 4 GHz
 C $\rightarrow$ 4 to 8 GHz
 X $\rightarrow$ 8 to 12 GHz
 Ku $\rightarrow$ 12 to 18 GHz
 K $\rightarrow$ 18 to 27 GHz
 Ka $\rightarrow$ 27 to 40 GHz

Standard Band Designation

ULF $\rightarrow$ 3 Hz to 30 Hz
 (ULTRA)
 ELF $\rightarrow$ 30 Hz to 300 Hz
 (EXTRA)
 VF $\rightarrow$ 300 Hz to 3 KHz
 (Voice)
 VLF $\rightarrow$ 3 KHz to 30 KHz
 LF $\rightarrow$ 30 KHz to 300 KHz
 MF $\rightarrow$ 300 KHz to 3 MHz
 HF $\rightarrow$ 3 MHz to 30 MHz
 VHF $\rightarrow$ 30 MHz to 300 MHz
 UHF $\rightarrow$ 300 MHz to 3 GHz
 SHF $\rightarrow$ 3 GHz to 30 GHz
 EHF $\rightarrow$ 30 GHz to 300 GHz

ADVANTAGES:

- 1) Fed BW availability
- 2) Improved Directive Property
- 3) Fading effect and reliability
- 4) Power Requirement
- 5) Transparency Property of microwave.
- 6) Long distance comm. systems.

APPLICATIONS:

- 1) Telecommunication: TV, Telephone, space Comm, Telemetry, Railway Comm.
- 2) Radar: ATC (air Traffic Control), burglar alarm, Police speed Detection
- 3) Commercial and Industrial application

Use heat property of microwave:

- Microwave oven
- Biomedical app.
- Drying m/c
- Food processing Ind. like Heat frozen/refrigerated meats

SIMILARITIES :-

- 1) Wave attenuated in both.
- 2) Wave reflected at the end of WG & TL.
- 3) Reflected wave can be eliminated by proper impedance matching.
- 4) Standing wave pattern is the result when both incident and reflected waves are present.

DISSIMILARITIES :-

- 1) There is a cut off value for the freq. of transmission depending upon the dimension and shape of WG. so WG is HPF as only waves having freq greater than f_c will be propagated.
- 2) WG is a one conductor transmission system.
- 3) In waveguide, wave impedance which is analogous to char. impedance Z_0 of TL.
- 4) The WG follows the field theory while the TL follows the circuit theory.



Wave Propagation (Wave Equations) (Helmholtz's Eqⁿ)

For the propagation of wave, all field vectors should vary w.r.t time and it requires harmonics in nature to propagate the wave forward.

Electric field (E) should contain harmonics.
Wave propagⁿ

so $E = E_0 e^{j\omega t}$ ①

where, $E_0 = \text{Max}^m$ value of EF intensity

$\omega = 2\pi f$, (Angular frequency of harmonics variations)

• Differentiating eqⁿ ①;

$$\frac{\partial E}{\partial t} = \underline{E_0 e^{j\omega t}} \cdot j\omega = E \cdot j\omega$$

$$\frac{\partial E}{\partial t} = j\omega E \quad \left[\frac{\partial}{\partial t} = j\omega \right] \text{ --- ②}$$

• Again taking differentiation;

$$\frac{\partial^2 E}{\partial t^2} = E_0 e^{j\omega t} \cdot (j\omega)^2$$

$$\frac{\partial^2 E}{\partial t^2} = -\omega^2 E_0 e^{j\omega t} = -\omega^2 \cdot E$$

$$\left[\frac{\partial^2}{\partial t^2} = -\omega^2 \right] \text{ --- ③}$$

• Now let us consider a medium which does not contain any free charges and is also non-conducting, ex. air or free space.

( $\sigma = 0$)
↓ Resistivity ↘ Conductivity

- From Maxwell's 1st equation; -

$$\nabla \times H = J + \frac{\partial D}{\partial t} = \sigma E + \frac{\partial (\epsilon E)}{\partial t}$$

for free space;

$$\nabla \times H = 0 + \epsilon \frac{\partial E}{\partial t} = \epsilon \cdot j\omega E$$

$$[\text{from Eqn } (2); \frac{\partial}{\partial t} = j\omega]$$

$$\nabla \times H = j\omega \epsilon E \quad \text{--- (4)}$$

- From Maxwell's 2nd eqn: -

$$\nabla \times E = -\frac{\partial B}{\partial t} = -\frac{\partial (\mu H)}{\partial t} = -j\omega \mu H$$

$$[\text{from (2); } \frac{\partial}{\partial t} = j\omega]$$

$$\nabla \times E = -j\omega \mu H \quad \text{--- (5)}$$

- Taking Curl of $\nabla \times E$; is - $\nabla \times (\nabla \times E)$

$$\nabla \times (\nabla \times E) = \nabla \times (-j\omega \mu H) \quad \text{--- (from (5))}$$

$$\nabla (\nabla \cdot E) - \nabla^2 E = -j\omega \mu (\nabla \times H)$$

$$= -j\omega \mu (j\omega \epsilon E) \quad \text{--- (from (4))}$$

$$\nabla (\nabla \cdot E) - \nabla^2 E = \omega^2 \mu \epsilon E \quad \text{--- (6)}$$

- From Maxwell's 3rd equation; -

$$\nabla \cdot D = \rho = 0 \quad (\text{for free space})$$

$$\nabla \cdot \epsilon E = 0$$

$$\epsilon (\nabla \cdot E) = 0$$

$$\text{But } \epsilon = \epsilon_0 \epsilon_r \neq 0$$

$$\text{So } \nabla \cdot E = 0$$

$$\text{From eqn (6); } -\nabla^2 E = \omega^2 \mu \epsilon E$$

$$\nabla^2 E = -\omega^2 \mu \epsilon E \quad \text{--- (7)}$$

Similarly for magnetic field Intensity;

$$\nabla^2 H = -\omega^2 \mu \epsilon H \quad \text{--- (8)}$$

Eqⁿ (7) and (8) are known as Helmholtz's equation, which are the wave equation to propagate in free space.

From (7) and (8);

If consider $E \rightarrow \hat{a}_x$ (x dirⁿ)

$M \rightarrow \hat{a}_y$ (y dirⁿ)

(Propagation dirⁿ) $\nabla \rightarrow \hat{a}_z$ (z direction)

As; $E \perp H \perp \text{Propa. Direction.}$

So we can modified eqⁿ (7) & (8) in unidirectional form;

$$\frac{\partial^2 E_z}{\partial z^2} = -\omega^2 \mu \epsilon E_z \quad \text{--- (9)}$$

constant
Propagation.

$$\frac{\partial^2 H_y}{\partial z^2} = -\omega^2 \mu \epsilon H_y \quad \text{--- (10)}$$

Propagation
constant

Solving the differential eqⁿ (9) + (10);

$$E(z)_x = (C_1 e^{-\gamma z} + C_2 e^{\gamma z}) a_x \quad \text{--- (11)}$$

indicate field direction
propagⁿ direction

$$H(z)_y = (C_1 e^{-\gamma z} + C_2 e^{\gamma z}) a_y \quad \text{--- (12)}$$

From eqⁿ (11), (12) to propagate wave in z-dirⁿ
General eqⁿ

$$E(z, t)_x = E_0 e^{j\omega t} e^{-\gamma z} a_x \rightarrow \text{field direction}$$

$$H(z, t)_y = H_0 e^{j\omega t} e^{-\gamma z} a_y \rightarrow \text{direction}$$

Wave Propagation (Wave Equation) (Helmholtz's Eqⁿ):

SIGNIFICANT: →

By the helmholtz's eqⁿ, we can easily understand that if source is time ~~domain~~ harmonic at one end then the effect is space harmonic propagation in the medium.

By the eqⁿ $\nabla^2 E = -\omega^2 \mu \epsilon E$ — (7) $\nabla^2 H =$

If $E \rightarrow x\text{-direction}$
 $M \rightarrow y\text{-direction}$
 Propagⁿ (∇) $\rightarrow x\text{-direction}$ } $E \perp H \perp PD$

~~to~~ eqⁿ from (1), (11), (12) eqⁿ;

Need to propagate any wave only a medium is not necessary but ~~also~~ surrounding (source) is also ~~also~~ required. like loop not Rod.



→ TEM :- All three axes are transverse to each other in space.

$\vec{E}$ along x , $\vec{H}$ along y , and $z \rightarrow$ propagation direction

Poynting vector $\vec{P} = \vec{E} \times \vec{H}$
(W/m²)

TEM wave are char. by ; $E_z = H_z = 0$

→ TM :- $H_z = 0$; $E_z \neq 0$

→ TE :- $H_z \neq 0$; $E_z = 0$

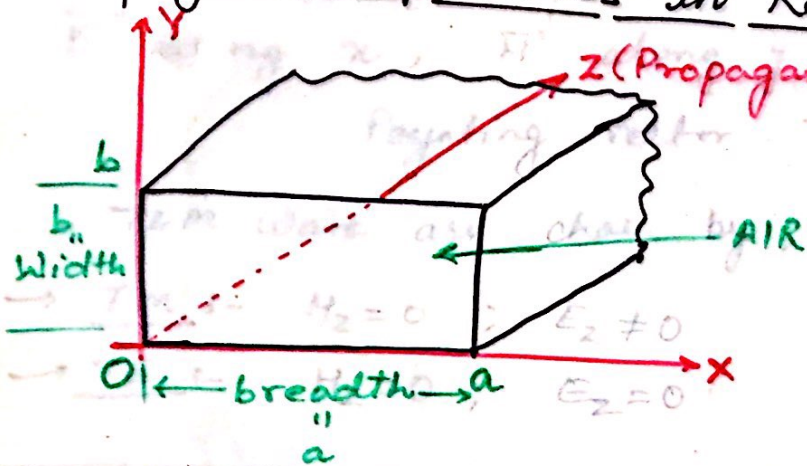
Waveguides :-

- At freq more than 3GHz, losses in the TLs and cables become significant. due to losses that occur in the dielectric needed to support the conductor.
- Metallic tube used to transmit EM waves at these freq.
- WG consists of a hollow metallic tube of rectangular or circular shape used to guide an EM waves by successive reflections from the inner walls of the tube.
- Thus no power is lost thru. radiation and dielectric loss is also neg. as WG are normally air filled.



RECTANGULAR WAVEGUIDE

Propagation of Waves in Rectangular Waveguide: →



As we know that; propagation direction is perpendicular to the cross product of E and M direction. ($E \perp M$ for propagation)

$E \text{ dir} \times M \text{ dir} = \text{Propagation Direction}$

Consider; $\downarrow$ $\downarrow$ $\downarrow$
 $(\hat{x}) \times (\hat{y}) = (\hat{z})$
 axis axis axis

- For Helmholtz equation; (TE and TM waves are rep. by —)

$$\left\{ \begin{array}{l} \nabla^2 H_z = -\omega^2 \mu \epsilon H_z \quad \text{TE mode} \\ \nabla^2 E_z = -\omega^2 \mu \epsilon E_z \quad \text{TM mode} \end{array} \right. \quad \left[\begin{array}{l} E_z = 0 \\ H_z = 0 \end{array} \right]$$

Taking z as a Propagation direction.

- $\nabla = \frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}$. (Rectangular Co-ordinate)

- Expanding $\nabla^2 E_z$ in rect. co-ordinate: →

$$\frac{\partial^2}{\partial x^2} E_z + \frac{\partial^2}{\partial y^2} E_z + \frac{\partial^2}{\partial z^2} E_z = -\omega^2 \mu \epsilon E_z$$



As wave is propagating in z-dirⁿ so there will be a propagation constant γ .

$$\text{so } \frac{\partial^2}{\partial z^2} = \gamma^2 \quad \text{--- (1)}$$

$$\bullet \frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + \gamma^2 E_z = -\omega^2 \mu \epsilon E_z$$

$$\frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + (\gamma^2 + \omega^2 \mu \epsilon) E_z = 0$$

$$\bullet \boxed{\gamma^2 + \omega^2 \mu \epsilon = k^2} \rightarrow \text{consider as a constant.}$$

$$\frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + k^2 E_z = 0 \quad (\text{for TM}) \quad \text{--- (2)}$$

$$\text{similarly; } \frac{\partial^2 H_z}{\partial x^2} + \frac{\partial^2 H_z}{\partial y^2} + k^2 H_z = 0 \quad (\text{for TE}) \quad \text{--- (3)}$$

• By solving (2)nd and (3)rd eqⁿ, we get E_z, H_z .

• By using Maxwell's eqⁿ, we can easily solve the other components also [E_z, H_x, E_y, H_y]

• From Maxwell's 1st eqⁿ;

$$\nabla \times \mathbf{H} = j\omega \epsilon \mathbf{E}$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix} = j\omega \epsilon [\hat{i} E_x + \hat{j} E_y + \hat{k} E_z]$$

($\frac{\partial}{\partial z} = -\gamma$ constant operator)

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & -\gamma \\ H_x & H_y & H_z \end{vmatrix} = j\omega \epsilon [\hat{i} E_x + \hat{j} E_y + \hat{k} E_z].$$



After Expanding;

$$\frac{\partial H_z}{\partial y} + \gamma H_y = j\omega \epsilon E_x \quad \text{--- (4)}$$

$$\frac{\partial H_z}{\partial x} + \gamma H_x = -j\omega \epsilon E_y \quad \text{--- (5)}$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = j\omega \epsilon E_z \quad \text{--- (6)}$$

Similarly for $\nabla \times E = -j\omega \mu H$

get $\frac{\partial E_z}{\partial y} + \gamma E_y = -j\omega \mu H_x \quad \text{--- (7)}$

$$\frac{\partial E_z}{\partial x} + \gamma E_x = +j\omega \mu H_y \quad \text{--- (8)}$$

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = -j\omega \mu H_z \quad \text{--- (9)}$$

→ From (4) and (8) eqⁿ; eliminate H_y , then;

$$H_y = \frac{1}{j\omega \mu} \frac{\partial E_z}{\partial x} + \frac{\gamma}{j\omega \mu} E_x \quad \text{--- (8th eqⁿ)}$$

Subs. in (4)th eqⁿ;

$$\frac{\partial H_z}{\partial y} + \frac{\gamma}{j\omega \mu} \frac{\partial E_z}{\partial x} + \frac{\gamma^2}{j\omega \mu} E_x = j\omega \epsilon E_x$$

$$E_x \left[j\omega \epsilon - \frac{\gamma^2}{j\omega \mu} \right] = \frac{\gamma}{j\omega \mu} \frac{\partial E_z}{\partial x} + \frac{\partial H_z}{\partial y}$$

Multiplied by $j\omega \mu$;

$$E_x [-\omega^2 \mu \epsilon - \gamma^2] = \gamma \frac{\partial E_z}{\partial x} + j\omega \mu \frac{\partial H_z}{\partial y}$$

$$E_x [-(\gamma^2 + \omega^2 \mu \epsilon)] = \gamma \frac{\partial E_z}{\partial x} + j\omega \mu \frac{\partial H_z}{\partial y}$$

$\parallel$
 h^2 (constant)

Dividing by $(-h^2)$;

$$E_x = -\frac{\gamma}{h^2} \frac{\partial E_z}{\partial x} - \frac{j\omega \mu}{h^2} \frac{\partial H_z}{\partial y} \quad \text{--- (10)}$$

Similarly; $E_y = -\frac{\gamma}{k^2} \frac{\partial E_z}{\partial y} + \frac{j\omega\mu}{k^2} \frac{\partial H_z}{\partial x}$ — (11)

$H_x = -\frac{\gamma}{k^2} \frac{\partial H_z}{\partial x} + \frac{j\omega\epsilon}{k^2} \frac{\partial E_z}{\partial y}$ — (12)

$H_y = -\frac{\gamma}{k^2} \frac{\partial H_z}{\partial y} - \frac{j\omega\epsilon}{k^2} \frac{\partial E_z}{\partial x}$ — (13)

These are the field component eqⁿs within rectangular waveguide.

Propagation of TEM modes :→

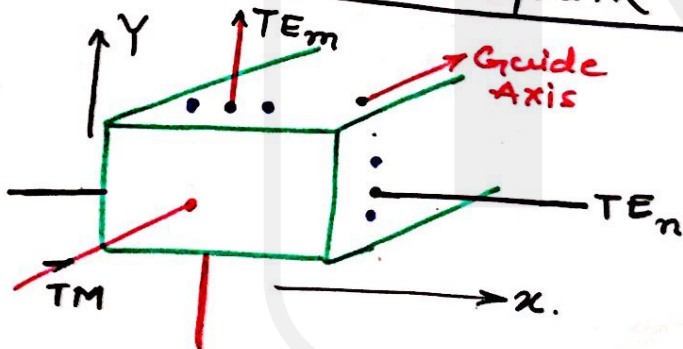
for TEM mode; $E_z = 0$ and $H_z = 0$

Substituting in above eqⁿ (10), (11), (12), (13)

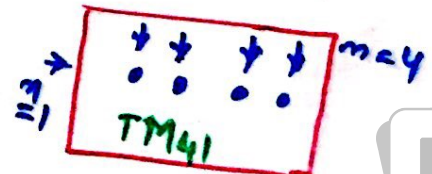
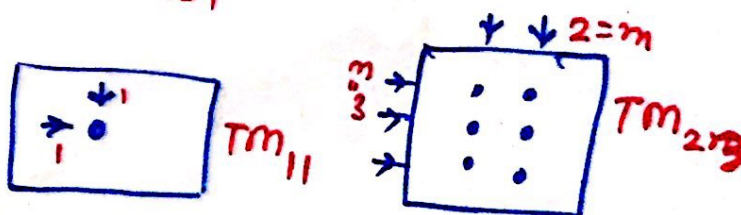
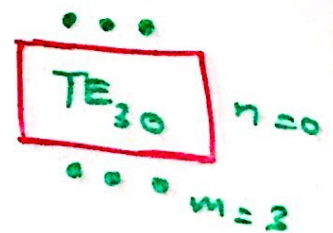
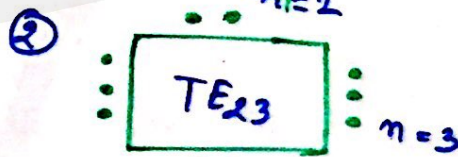
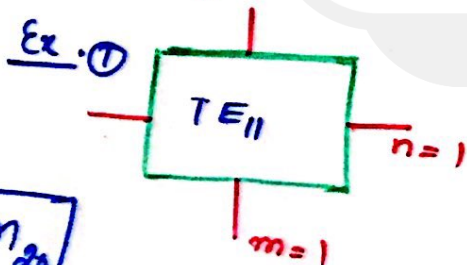
all the field components vanish and hence

"TEM mode in Rectangular Waveguide does not exist"

* MODES IN RECTANGULAR WG :→



The no of out of phase feed connections decides the no of m's and n's.



Que:- 5(a):

→ Propagation of TM waves in rectangular WG:- →

• If propagation is in direction 'z'.

• For TM wave; $H_z = 0$; $E_z \neq 0$

• From propagⁿ wave eqⁿ (2) in RWG;

$$\frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + h^2 E_z = 0 \quad \text{--- (1)}$$

To solve the different field components E_x, E_y, H_x, H_y we have to use SEPARATION OF VARIABLES METHOD.

Let us consider $E_z = X Y \rightarrow$ Pure y components.
 $\downarrow$
 Pure x components

• where x and y are the independent variables,

$$\frac{\partial^2 E_z}{\partial x^2} = \frac{\partial^2 (X Y)}{\partial x^2} = Y \frac{\partial^2 X}{\partial x^2}$$

$$\frac{\partial^2 E_z}{\partial y^2} = \frac{\partial^2 (X Y)}{\partial y^2} = X \frac{\partial^2 Y}{\partial y^2}$$

• From (1);

$$Y \frac{\partial^2 X}{\partial x^2} + X \frac{\partial^2 Y}{\partial y^2} + h^2 X Y = 0$$

• Dividing by XY ;

$$\frac{1}{X} \frac{\partial^2 X}{\partial x^2} + \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} + h^2 = 0$$

↓
 Pure funcⁿ of x -only

↓
 Pure funcⁿ of y -only.

(2)

50
bag
rol
T
m

These two funcⁿs are constants.

let $\frac{1}{x} \frac{\partial^2 X}{\partial x^2} = -B^2$

$$\frac{\partial^2 X}{\partial x^2} = -B^2 X \quad \text{--- (3)}$$

let $\frac{1}{y} \frac{\partial^2 Y}{\partial y^2} = -A^2$; $\frac{\partial^2 Y}{\partial y^2} = -A^2 Y \quad \text{--- (4)}$

From eqⁿ (2) ; substitute (3) and (4) in eqⁿ;

$$-B^2 - A^2 + k^2 = 0$$

$$k^2 = A^2 + B^2 \quad \text{--- (5)}$$

Eqⁿ (3)rd and (4)th are partial differential eqⁿ so solving as 2nd order differential eqⁿ;

$$X = C_1 \cos Bx + C_2 \sin Bx. \quad \text{--- (6)}$$

$$Y = C_3 \cos Ay + C_4 \sin Ay \quad \text{--- (7)}$$

where C_1, C_2, C_3 and C_4 are constants.

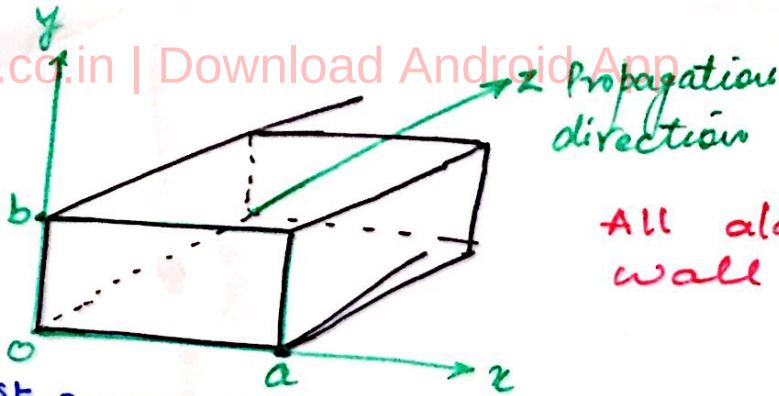
Get the solution of $E_z = X \cdot Y$ By putting eqⁿ (6)th and (7)th.

Boundary Condition exist in RWG as at the surface RWG behaves as short circuit and $E_z = 0$, all along the wall of RWG. So for 4 Boundaries there will be 4 Boundary Conditions.

$$E_z = (C_1 \cos Bx + C_2 \sin Bx)(C_3 \cos Ay + C_4 \sin Ay)$$



Pro
If
For
from



Walls are perfect
conducting sheet
so $\rightarrow$
All along the boundary
wall $E_z = 0$

1st BOUNDARY CONDITION: [Bottom wall]; i.e. $y=0$
 $E_z = 0$
 $x \rightarrow 0 \text{ to } a$

2nd BCondition:- [Left side wall]; $x=0$
 $E_z = 0$
 $y \rightarrow 0 \text{ to } b$

3rd :- [Top plane wall] i.e., $y=b$
 $E_z = 0$
 $x \rightarrow 0 \text{ to } a$

4th :- [Right side wall] i.e., $x=a$
 $E_z = 0$
 $y \rightarrow 0 \text{ to } b$

• From 8th eqⁿ; After applying 1st BC;
 $E_z = 0$ at $y=0$ and $x \rightarrow 0 \text{ to } a$

$$E_z = (C_1 \cos Bx + C_2 \sin Bx) (C_3 \cos Ay + C_4 \sin Ay)$$

$$0 = (C_1 \cos Bx + C_2 \sin Bx) \cdot C_3$$

As; $C_1 \cos Bx + C_2 \sin Bx \neq 0$ [x is varying]
so $C_3 = 0$ ——— (9a)

so eqⁿ 8th reduces to;

$$E_z = (C_1 \cos Bx + C_2 \sin Bx) (C_4 \sin Ay)$$

• Similarly after applying 2nd B.C; and
condition comes from 1st BC (9a) eqⁿ;

$$0 = C_1(C_4 \sin A_y)$$

As $C_4 \sin A_y \neq 0$ or $C_4 \neq 0$ (as C_4 is varying)

$$\text{so } C_1 = 0 \quad \text{--- (9c)}$$

so eqⁿ (9) reduces to;

$$E_2 = (C_2 \sin Bx)(C_4 \sin Ay) \quad \text{--- (10)}$$

• After applying 3rd B.C ; and (9c) ;

$$0 = \underbrace{C_2 \sin Bx}_{\substack{\downarrow \\ x \text{ is varying} \\ \text{can't be} \\ \text{zero}}} \cdot C_4 \sin Ab ; \quad \begin{matrix} y=b \\ \downarrow \\ x \rightarrow 0 \text{ to } a \\ \text{constant} \end{matrix}$$

$$\text{so } 0 = C_4 \sin Ab ; \quad \sin Ab = 0$$

$$\text{so } \sin Ab = \sin n\pi = 0 \quad \begin{matrix} Ab = \text{multiple of } \pi \\ = n\pi \end{matrix}$$

$$Ab = n\pi$$

$$\boxed{A = \frac{n\pi}{b}} \quad \therefore \text{where } n = 0, 1, 2, \dots \quad \text{--- (9d)}$$

• After applying 4th B.C ;

$$0 = C_2 \sin Ba \cdot \underbrace{C_4 \sin Ay}_{\substack{\downarrow \\ \text{can't zero}}} ; \quad \begin{matrix} x=a \\ y \rightarrow 0 \text{ to } b \end{matrix}$$

$$0 = C_2 \sin Ba ; \quad \sin Ba = 0 = \sin m\pi$$

$$Ba = m\pi$$

$$\boxed{B = \frac{m\pi}{a}} \quad \text{--- (9e)}$$

5(b): TM_{01} ;

$$H_x = \frac{j\omega\epsilon}{h^2} c \left(\frac{n\pi}{b}\right) \sin\left(\frac{m\pi}{a}\right) z \cos\left(\frac{n\pi}{b}\right) y \rightarrow 0 \cdot e^{-j\omega t - \gamma z}$$

$$m=0 ; n=1$$

Similarly; $H_x = 0$
 $H_y = 0$

$$E_x = 0 ; E_y = 0$$

All the fields components vanished so TM_{01} doesn't exist.

Similarly; $TM_{10} \rightarrow$ known as even-scent modes which are physically not possible to make.



$$\therefore \text{from eq}^n (4); E_z = C_2 C_4 \sin\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \cdot e^{-\gamma z} \cdot e^{j\omega t}$$

where; $e^{-\gamma z}$ = propagation along 'z' direction
 $e^{j\omega t}$ = sinusoidal variation w.r.t 't'.

$C_2 C_4 = C$ = other constant

$$E_z = C \sin\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \cdot e^{-\gamma z} \cdot e^{j\omega t}$$

from the previous general eqⁿ to propagate any wave.

• from the wave propagⁿ in RWG general eqⁿ;

$$\# E_x = -\frac{\gamma}{h^2} \frac{\partial E_z}{\partial x} - \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial y}$$

(so for a TM wave $H_z = 0$)

$$E_x = -\frac{\gamma}{h^2} \frac{\partial E_z}{\partial x}$$

To differentiate E_z ;

$$E_x = -\frac{\gamma}{h^2} C \left(\frac{m\pi}{a}\right) \cos\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \cdot e^{j\omega t - \gamma z}$$

(all 4 eq's shows that TM0n and TMnm not exist.)

$$E_y = -\frac{\gamma}{h^2} \frac{\partial E_z}{\partial y} + \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial x} \quad (\text{TM}; H_z = 0)$$

$$E_y = -\frac{\gamma}{h^2} C \left(\frac{n\pi}{b}\right) \sin\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y \cdot e^{j\omega t - \gamma z}$$

$$H_x = -\frac{\gamma}{h^2} \frac{\partial H_z}{\partial x} + \frac{j\omega\epsilon}{h^2} \frac{\partial E_z}{\partial y}$$

$$H_x = \frac{j\omega\epsilon}{h^2} C \left(\frac{n\pi}{b}\right) \sin\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y \cdot e^{j\omega t - \gamma z}$$

$$H_y = -\frac{\gamma}{h^2} \frac{\partial H_z}{\partial y} - \frac{j\omega\epsilon}{h^2} \frac{\partial E_z}{\partial x} = \frac{j\omega\epsilon}{h^2} C \left(\frac{m\pi}{a}\right) \cos\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \cdot e^{j\omega t - \gamma z}$$

$$E(x, y, z, t)_z = \underset{\substack{\parallel \\ E_{z0}}}{C} \sin\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y e^{-r_z} e^{j\omega t} \hat{a}_z$$

in EF $\begin{cases} x \text{ to } z \rightarrow \text{Normal} \rightarrow \text{sign sin} & (E \text{ to } E) \\ y \text{ to } z \rightarrow \text{Normal} \rightarrow \text{sin} \end{cases}$

$$1) E(x, y, z, t)_y = \underset{\substack{\text{tang} \\ \text{Normal}}}{C \left(\frac{m}{b}\right)} \sin\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y e^{-r_z} e^{j\omega t} \hat{a}_y$$

$(E \xrightarrow{\text{sin}} EF) \{ y \rightarrow y \rightarrow \text{tangential} \rightarrow \text{taking differentiating}$

$$2) H(x, y, z, t)_y = \underset{\substack{\text{tang} \\ \text{Normal}}}{C \left(\frac{m}{a}\right)} \cos\left(\frac{m\pi}{a}\right)x \cdot \sin\left(\frac{n\pi}{b}\right)y$$

$$3) E(x, y, z, t)_x = \underset{\substack{\text{Normal} \\ \text{tang}}}{C \left(\frac{m}{a}\right)} \cos\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y$$

$$4) H(x, y, z, t)_x = \underset{\substack{\text{Normal} \\ \text{tang} \parallel \text{Normal}}}{C \left(\frac{m}{a}\right)} \sin\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y$$

$$H(x, y, z, t)_z = C \cdot \cos\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y e^{-r_z} e^{j\omega t} \hat{a}_z$$

$$1) H(x, y, z, t)_x = \underset{\substack{\text{tang} \\ \text{N}}}{C \left(\frac{m}{b}\right)} \sin\left(\frac{m\pi}{a}\right)x \frac{\sin\left(\frac{n\pi}{b}\right)}{\cos\left(\frac{n\pi}{b}\right)}y$$

$(H \text{ to } H) \text{ tang} \rightarrow \text{taking differentiation}$
 $(H \text{ to } H) \text{ Normal} \rightarrow \text{No change.}$

$$2) E(x, y, z, t)_x = \underset{\substack{\text{N} \\ \text{T}}}{C \left(\frac{n}{b}\right)} \cos\left(\frac{m\pi}{a}\right)x \cdot \sin\left(\frac{n\pi}{b}\right)y$$

Dominant, Degenerated and Evanescent modes:-

→ As we seen earlier that TM_{m0} and TM_{0n} modes in TM modes do not exist so these modes known as evanescent modes.

Where TE_{m0} or TE_{0n} exists.

→ Dominant mode:- Having lowest cut off freq. known as dominant mode.

$$f_c = \frac{c}{2} \left[\left(\frac{m}{a} \right)^2 + \left(\frac{n}{b} \right)^2 \right]^{1/2}$$

for TE Modes : ① TE_{01} , ② TE_{10} , ③ TE_{11}

$$f_c = \frac{c}{2b}$$

$$f_c = \frac{c}{2a}$$

$$f_c = \frac{c}{2} \left[\frac{1}{a^2} + \frac{1}{b^2} \right]^{1/2}$$

①st and ②nd cut off freq. is lowest than TE_{11} depending dimension of 'a' and 'b'.

If $a > b$ then TE_{01} (dominant mode)

If $a < b$ then TE_{10} (dominant mode)

for TM modes : ① TM_{11} > ② TM_{12} > ③ TM_{21}

$$f_{c11} = \frac{c}{2} \left[\frac{1}{a^2} + \frac{1}{b^2} \right]^{1/2} \quad f_{c12} = \frac{c}{2} \left[\left(\frac{1}{a} \right)^2 + \frac{4}{b^2} \right]^{1/2}$$

The Lowest cut off freq. in TM modes is TM_{11} , so the dominant mode is TM_{11}

→ Degenerated mode:- When two different modes having same cut off freq. is known as degenerated mode.

Ex. for TE_{32} and TM_{32} cut off freq. is same; $\frac{c}{2} \left[\frac{9}{a^2} + \frac{4}{b^2} \right]^{1/2}$

TE_{32} and $TM_{32} \rightarrow$ degenerated mode

TE_{12} and $TM_{12} \rightarrow$ " etc.

Ex. TM_{32} or TE_{23} also degenerated mode if $\frac{a=b}{\downarrow}$
(square waveguide)

→ TM modes in RWG : → (TM_{mn})

1) TM₀₀ mode :- $m=n=0$; By putting E_x, E_y, H_x, H_y vanish, so can't exist.

2) TM₀₁ mode :- Not exist

3) TM₁₀, TM₂₀, TM₀₂ → All are known as even modes

4) TM₁₁ :- All the H field component will exist so TM₁₁ will exist and all higher modes do exist.

*→ Cut off freq of RWG : →

We know that; $k^2 = \gamma^2 + \omega^2 \mu \epsilon = A^2 + B^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$

$$\gamma^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \omega^2 \mu \epsilon$$

$$\gamma = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \omega^2 \mu \epsilon} = \alpha + j\beta$$

$\downarrow$ Attenuation Constant $\downarrow$ phase const.

⇒ At lower freq;

$$\omega^2 \mu \epsilon < \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

then $\gamma = +ve$ only + real

so $\gamma = \alpha$, i.e., all components will be attenuated not propagated.

⇒ At higher freq; $\omega^2 \mu \epsilon > \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$

then $\gamma = -j\beta$ imaginary.

so $\gamma = \beta$ i.e., propagation just starts.

⇒ TD

1)

"freq at which γ just becomes zero is defined as cut off freq (threshold freq) f_c ."

→ As RWG passes high freq and stop low freq so it acts as HPF.

→ At exact cut off freq, the wave oscillates b/w the guide walls, is not propagate forward. It is resonating b/w the conducting walls.

→ At $f = f_c$; $\gamma = 0$ or $\omega = \omega_c = 2\pi f_c$

$$0 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \omega_c^2 \mu \epsilon$$

$$\omega_c^2 \mu \epsilon = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

$$f_c = \frac{1}{2\pi \sqrt{\mu \epsilon}} \left[\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]^{1/2}$$

$$\# \quad f_c = \frac{c}{2} \left[\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 \right]^{1/2}$$

$$\lambda_c = \frac{c}{f_c} = \frac{2ab}{\sqrt{m^2 b^2 + n^2 a^2}}$$

→ Phase velocity of RWG: $-(V_p)$:

We know that

$$\alpha + j\beta = \gamma = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \omega^2 \mu \epsilon} = j \sqrt{\omega^2 \mu \epsilon - \left[\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right]}$$

$$\beta = \sqrt{\omega^2 \mu \epsilon - \left[\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right]}$$

from ① above

$$\# \quad \sqrt{\beta} = \omega^2 \mu \epsilon - \omega_c^2 \mu \epsilon$$



$$\begin{aligned}
 v_p &= \frac{\omega}{\beta} = \frac{\omega}{\sqrt{\omega^2 \mu \epsilon - \omega_c^2 \mu \epsilon}} = \text{Phase velocity along the guide axis.} \\
 &= \frac{\omega}{\sqrt{\mu \epsilon} \sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2}} = \frac{c}{\sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2}} \quad [c = \sqrt{\mu \epsilon}]
 \end{aligned}$$

$$v_p = \frac{c}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} = \frac{c}{\sqrt{1 - \left(\frac{\lambda}{\lambda_c}\right)^2}} \quad \text{--- ①}$$

Significance Every freq. is choosing its freq. itself.

* v_p defined as the rate at which the wave changes its phase in terms of the guide wavelength.

Group velocity : $\rightarrow$ If there is modulation in the carrier, the modulation envelope actually travels at velocity slower than that of carrier alone which is slower than speed of light.

The velocity of modulation envelope is called the group velocity (v_g). This occurs when a modulated sig travels in a waveguide.

$$v_g = \frac{d\omega}{d\beta}$$

$$\beta = \sqrt{\omega^2 \mu \epsilon - \omega_c^2 \mu \epsilon}$$

$$\frac{d\beta}{d\omega} = \frac{1}{2} \frac{2\omega \mu \epsilon}{\sqrt{\omega^2 \mu \epsilon - \omega_c^2 \mu \epsilon}}$$

$$= \frac{\omega \mu \epsilon / \sqrt{\mu \epsilon}}{\omega \sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2}}$$

$$= \frac{\sqrt{\mu \epsilon}}{\sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2}}$$

$$\boxed{\frac{d\beta}{d\omega} = \frac{1}{c \sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2}}}$$

$$v_g = \frac{d\omega}{d\beta} = c \sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2} = c \sqrt{1 - \left(\frac{f_c}{f}\right)^2} = c \cos \theta$$

$$\boxed{v_g = c \cos \theta}$$

$$\bullet v_g < c$$

$$\boxed{v_p = \frac{c}{\cos \theta}}$$

$$\bullet v_p > c$$

$$\boxed{v_p \cdot v_g = c^2}$$

⇒ Similarly Topics has been covered -

- 1) Propagation of TE wave in RWG
- 2) TE modes in RWG



→ Power losses or Attenuation effect in Waveguide:

We know that: $\gamma = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \omega^2 \mu \epsilon} = \alpha + j\beta$

If $\omega^2 \mu \epsilon < \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$; $\omega < \omega_c$

and $\gamma = +ve$ and real = α

So all the field component will be attenuated at below cut off freq.

⇒ We know $\beta = \frac{2\pi}{\lambda_g}$; $\lambda_g = \frac{\lambda}{\cos \theta} = \frac{\lambda}{\sqrt{1 - (f_c/f)^2}}$

$$\beta = \frac{2\pi}{\lambda_g} = \frac{2\pi}{\lambda} \sqrt{1 - (f_c/f)^2} = j \frac{2\pi}{\lambda} \sqrt{(f_c/f)^2 - 1}$$

$$\beta = j \frac{2\pi}{\lambda} \cdot \frac{f_c}{f} \sqrt{1 - (f/f_c)^2} = j \frac{2\pi f_c}{c} \sqrt{1 - (f/f_c)^2}$$

⇒ As instead of propagation wave will be ($\beta = j\alpha$) attenuated.

$$\alpha = \frac{2\pi f_c}{c} \sqrt{1 - (f/f_c)^2}$$

In dB;

$$\alpha = \frac{54.6}{\lambda_c} \sqrt{1 - (f/f_c)^2} \text{ dB/length.}$$

⇒ actually this α works as stop band attenuation of the waveguide high pass filter.

for $f > f_c$; very low loss (α)

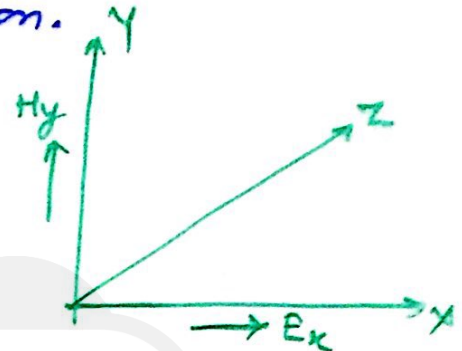
$f > f_c$; $\alpha \uparrow$; full reflection known as reflection loss.

Wave impedance (Z_z) in TM and TE waves :-
or (Z_g)

Z_g is the ratio of EF in one transverse direction to the strength of the MF along the other transverse direction.

$$Z_z = \frac{E_x}{H_y} = \frac{-E_y}{H_x}$$

$$\text{or } Z_z = \frac{\sqrt{E_x^2 + E_y^2}}{\sqrt{H_x^2 + H_y^2}}$$



1) For TM Wave in RWG;

$$Z_z = Z_{TM} = \frac{E_x}{H_y} = \frac{-\frac{\gamma}{k^2} \frac{\partial E_z}{\partial x} - \frac{j\omega\mu}{k^2} \frac{\partial H_z}{\partial y}}{-\frac{\gamma}{k^2} \frac{\partial H_z}{\partial y} - \frac{j\omega\epsilon}{k^2} \frac{\partial E_z}{\partial x}}$$

for TM ; $H_z = 0$ and $\gamma = j\beta$

$$Z_{TM} = \frac{-\frac{\gamma}{k^2} \frac{\partial E_z}{\partial x}}{-\frac{j\omega\epsilon}{k^2} \frac{\partial E_z}{\partial x}} = \frac{\gamma}{j\omega\epsilon} = \frac{j\beta}{j\omega\epsilon} = \frac{\beta}{\omega\epsilon}$$

$$(\beta = \sqrt{\omega^2\mu\epsilon - \omega_c^2\mu\epsilon})$$

$$Z_{TM} = \frac{\sqrt{\omega^2\mu\epsilon - \omega_c^2\mu\epsilon}}{\omega\epsilon} = \frac{\sqrt{\mu\epsilon} \sqrt{1 - \left(\frac{\omega_c^2}{\omega^2}\right)}}{\omega\epsilon}$$

$$= \sqrt{\frac{\mu}{\epsilon}} \sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2} = \eta \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

$$\boxed{\eta = \text{intrinsic Impedance} = \sqrt{\frac{\mu}{\epsilon}}}$$

$$\# \boxed{Z_{TM} = \eta \sqrt{1 - \left(\frac{f_c}{f}\right)^2}} \quad \text{Similarly;}$$

$$\bullet Z_{TM} < \eta \bullet$$

$$\bullet Z_{TE} > \eta \bullet$$

$$\boxed{Z_{TE} = \frac{\eta}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}}$$

$$\boxed{Z_{TM} \cdot Z_{TE} = \eta^2}$$

- TM_{m0} and TM_{0n} modes can't be physically connected they are even-scent modes.
- Ex. TM_{20} , TM_{02} ...

MODES

Wave Angle or Tilt Angle! $\rightarrow (\theta)$

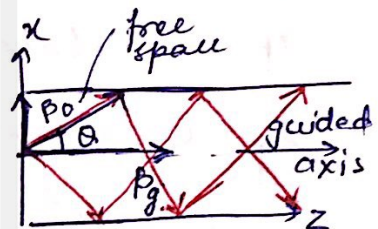
Taking 2-D RWG to understand this,

where β_g = guided wave phase constant

$$= \frac{2\pi}{\lambda_g}$$

and β_0 = free space phase constant $= \frac{2\pi}{\lambda_0}$

$$\beta_g = \beta_0 \cos \theta \quad ; \quad \frac{2\pi}{\lambda_g} = \frac{2\pi}{\lambda_0} \cos \theta$$



$$\cos \theta = \frac{\lambda_0}{\lambda_g} \quad ; \quad \boxed{\lambda_g = \frac{\lambda_0}{\cos \theta}}$$

~~$\cos \theta =$~~

$$f \lambda_g = \frac{f \lambda_0}{\cos \theta}$$

(Phase velo) $v_p = \frac{c}{\cos \theta}$ — (from eq. of v_p)

$\lambda_c = \frac{c}{f}$
 $\lambda_0 = \frac{c}{f}$
 $\lambda_g = \frac{c}{f}$

✓ Tilt angle
 $\cos \theta = \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$
 so $\sin \theta = \frac{f_c}{f}$
 ↑
 (from eq. of v_p)



→ Phase velocity, Group velocity, Guide Wavelength and Wave impedance : →

$$v_g = v_0 \sqrt{1 - (f_c/f)^2}$$

$$Z_{TE} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}} ; Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$$

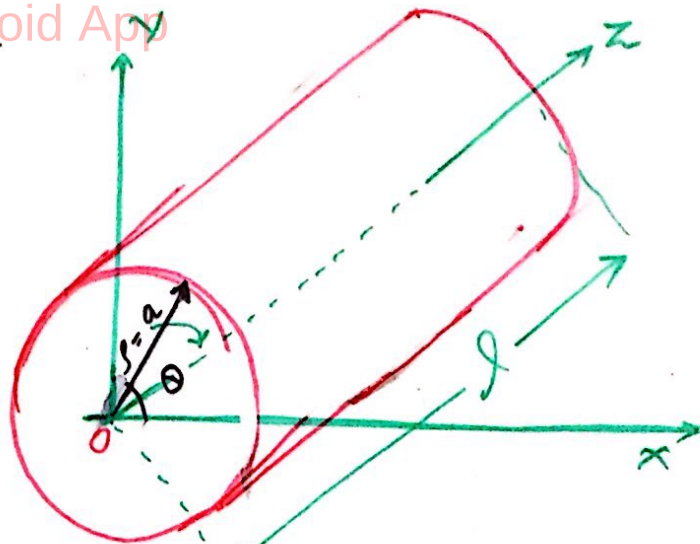
TEM mode :- NOT exist in CWG.

*(Advantage, Disadvantage and application)



CIRCULAR Waveguide:-

circular WG basically depends on 3 variable and circular co-ordinate geometry. (ρ, ϕ, z)



- Here ϕ varies from 0 to 2π , ρ varies 0 to a and z varies along z -axis.

• Propagation of TE waves in CWG:-

• For TE wave $E_z = 0$ and $H_z \neq 0$

• From Maxwell eqⁿ; $\nabla^2 H_z = -\omega^2 \mu \epsilon H_z$

• Expanding ∇ in circular coordinates;

$$\frac{\partial^2 H_z}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial H_z}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 H_z}{\partial \phi^2} + \frac{\partial^2 H_z}{\partial z^2} = -\omega^2 \mu \epsilon H_z$$

As we know; $\partial^2 / \partial z^2 = -k^2$ (an operation)

$$\frac{\partial^2 H_z}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial H_z}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 H_z}{\partial \phi^2} + (k^2 + \omega^2 \mu \epsilon) H_z = 0$$

• $k^2 + \omega^2 \mu \epsilon = h^2$ (we know)

$$\frac{\partial^2 H_z}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial H_z}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 H_z}{\partial \phi^2} + h^2 H_z = 0$$

• This differential eqⁿ can be solved by "VARIABLE'S SEPARATION" Method.

So let's consider $H_2 = P \cdot Q$.
 where $P =$ is a funcⁿ of ρ only
 $Q =$ " " " " ϕ only.

$$\text{So: } \frac{\partial^2(PQ)}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial(PQ)}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2(PQ)}{\partial \phi^2} + \hbar^2 PQ = 0$$

$$Q \frac{\partial^2 P}{\partial \rho^2} + \frac{Q}{\rho} \frac{\partial P}{\partial \rho} + \frac{P}{\rho^2} \frac{\partial^2 Q}{\partial \phi^2} + \hbar^2 PQ = 0$$

→ Multiplying throughout by $\frac{\rho^2}{PQ}$;

$$\frac{\rho^2}{P} \frac{d^2 P}{d\rho^2} + \frac{\rho}{P} \frac{dP}{d\rho} + \frac{1}{Q} \frac{d^2 Q}{d\phi^2} + \hbar^2 \rho^2 = 0$$

(1) (2) (3) (4)

all the (1) (2) (3) and (4) parts are the pure funcⁿ of ρ and ϕ only.

→ so let $\frac{1}{Q} \cdot \frac{d^2 Q}{d\phi^2} = -n^2$ (Pure funcⁿ of ϕ)

→ Divide by P : -

$$\left\{ \frac{\rho^2}{P} \frac{d^2 P}{d\rho^2} + \frac{\rho}{P} \frac{dP}{d\rho} + (\rho^2 \hbar^2 - n^2) P = 0 \right\}$$

By using Bessel funcⁿ, we can solve the value of P and Q :

$$P = C_n J_n(\rho \hbar)$$

$$Q = \underbrace{\sqrt{A_n^2 + B_n^2}}_{C'_1} \cos\left(\underbrace{n\phi}_{n\phi} + \tan^{-1} \frac{A_n}{B_n}\right)$$

$$H_2 = P \cdot Q = C_n J_n(\rho \hbar) C'_1 \cos n\phi$$

$$H_2 = C_0 J_n(\rho \hbar) \cos n\phi \quad (C_n C'_1 = C_0)$$

A_z in RWG, in CWG we can calculate all field component $E_\rho, E_\phi, E_z, H_\rho, H_\phi, H_z$ which are analogous to $E_x, E_y, 0, H_x, H_y, 0$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $E_x \quad E_y \quad 0 \quad H_x \quad H_y \quad 0$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 TEMode TMmode

as in RWG; (TE mode); in CWG (TE mode);

$$E_x = -\frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial y}$$

$$E_y = \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial x}$$

$$H_x = -\frac{\gamma}{h^2} \frac{\partial H_z}{\partial x}$$

$$H_y = -\frac{\gamma}{h^2} \frac{\partial H_z}{\partial y}$$

$$E_\rho = -\frac{j\omega\mu}{h^2} \frac{1}{\rho} \frac{\partial H_z}{\partial \phi}$$

$$E_\phi = \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial \rho}$$

$$H_\rho = -\frac{\gamma}{h^2} \frac{\partial H_z}{\partial \rho}$$

$$H_\phi = -\frac{\gamma}{h^2} \frac{1}{\rho} \frac{\partial H_z}{\partial \phi}$$

$$H_z = G_0 J_n(\rho h) \cos n\phi e^{-\gamma z}$$

where $h^2 = \gamma^2 + \omega^2\mu\epsilon$

- Propagation of TM wave in CWG (Assignment)
 P_{nm} values for TM modes in CWG;

$\begin{matrix} m \\ n \end{matrix}$	1	2	3
0	2.405	5.520	8.645
1	3.832	7.106	10.173
2	5.135	8.417	11.620
3	6.380	9.761	13.015

So for general eqⁿ, where consider a sinusoidal variation along 'z';

$$H_z = C_0 J_n(\rho h) \cos n\phi \cdot e^{-\gamma z} \quad \text{--- (1)}$$

Boundary Condition: → We know all along the surface area of the circular WG at $\rho = a$
 $E_\phi = 0$ for all values of ϕ varying b/w 0 to 2π .

i.e., $\left. \frac{\partial H_z}{\partial \rho} \right|_{\rho=a} = 0$

From the 1st eqⁿ; $C_0 J'_n(\rho h) \cos n\phi \cdot e^{-\gamma z} \Big|_{\rho=a} = 0$

$$J'_n(ah) = 0$$

→ Bessel funcⁿ $J'_n(ah)$ denotes prime differentia. with resp. to ah , so m^{th} root of this eqⁿ is denoted by P'_{nm} , which are the eigen values;

$$\{ P'_{nm} = ah \} \text{ for } TE_{nm}$$

$$h = \frac{P'_{nm}}{a} \rightarrow \text{this rep. all}$$

possible soluⁿ of H_z for TE_{nm} in CWG.

$n \backslash m$	1	2	3
0	3.832	7.016	10.173
1	1.841	5.331	8.536
2	3.054	6.706	9.969
3	4.201	8.015	11.346

P'_{nm} values for TE_{nm} modes in CWG

⇒ cut-off wavelength in circular waveguide

$$\left[\begin{array}{l} \text{for TE; } \lambda_c = \frac{2\pi a}{P'_{nm}} \\ \text{for TM; } \lambda_c = \frac{2\pi a}{P_{nm}} \end{array} \right] \quad \text{--- (2)}$$

⇒ Dominant mode:- Having lowest cut off freq or highest wavelength cut off:

∴ from 2nd eqⁿ; To be λ_c highest, P'_{nm} lowest
so for $n=1, m=1$, $P'_{nm} = 1.841$ (lowest in Table)

so TE₁₁ is dominant mode in CWG.

$$\lambda_c = \frac{2\pi a}{1.841}$$

∴ from eqⁿ 3rd; To be λ_c best, P_{nm} (best)
so for $n=0, m=1$, $P_{nm} = 2.405$ (best)

so TM₀₁ is dominant mode in CWG.

$$\lambda_c = \frac{2\pi a}{2.405}$$

⇒ Degenerated Mode:- from P'_{nm} and P_{nm} Table; we can see that $P'_{0m} = P_{1m}$
Hence TE_{0m} and TM_{1m} are degenerated in uniform CWG.

→ INTRODUCTION TO PLANAR TRANSMISSION LINES: →

- The conventional open wire TL are not suitable for MW transmission b'coz of radiation loss.
- Due to bulky structure of co-axial.
- At MW freq, the following TL are employed:-

(1) Single conductor lines (WGL)

(a) RWGL (b) CWGL (c) coplanar lines etc.

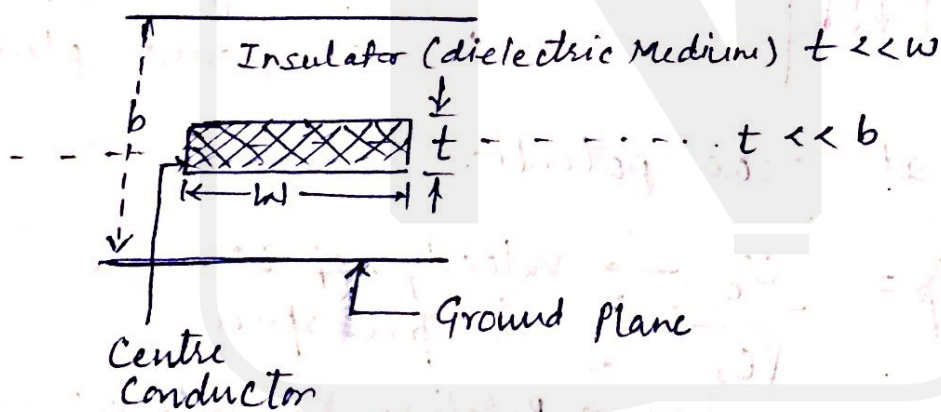
(2) Multiconductor lines (Planar TL)

(a) coaxial lines (b) strip lines
(c) Microstrip lines (d) parallel strip lines
(e) coplanar lines (f) coupled strip line.

(3) open-boundary structures:

(a) dielectric rods. (b) open waveguide, etc.

→ The Strip Line :→ It is a three conductor TL used at frequencies b/w 100MHz to 100GHz .



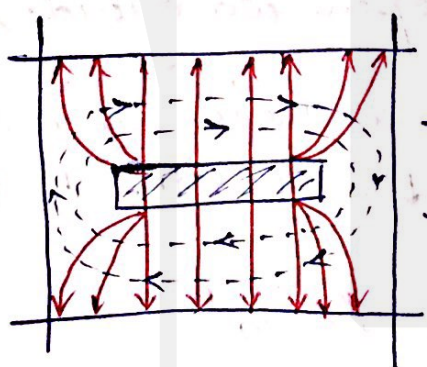
- SL consists of a central thin conducting strip of width and thickness t . There are 2 ground plates. There is a homogeneous low loss dielectric b/w the two ground planes. The centre conductor is embedded in dielectric medium.

(2)

→ strip configuration may be either symmetric or asymmetrical. The symmetrical structure is called strip line while the asymmetric arrangement is denoted as microstrip.

→ The voltage is applied b/w centre strip and pair of ground planes. The current flows down the centre strip and returns via ground planes. ~~Although~~ Although the structure is open at both the end, even it is non radiating TL.

→ Dominant mode for $SL \rightarrow \underline{TEM \text{ mode}}$



→ The field lines are concentrated near the centre strip and decay exponentially towards ground planes

→ Phase velocity $v_p = \frac{c}{\sqrt{\epsilon_r}}$

c → velocity of EM wave in free space
 ϵ_r → Relative permittivity

$$\xi Z_0 = \frac{1}{v} c \quad \text{unit capacitance of unit length of line}$$

Guide wavelength: $\rightarrow (\lambda_g) = 2\pi / \beta_g$.

Defined as the distance travelled by the wave in order to undergo a phase shift of 2π radian. It is related to the phase constant.

λ_g is slightly different from λ_0 (free space WL) and λ_c (cut of WL) by: $\lambda_g = \lambda_0 / \cos \theta$.

or so $\frac{1}{\lambda_g^2} = \frac{1}{\lambda_0^2} - \frac{1}{\lambda_c^2}$

$$\lambda_g = \frac{\lambda_0}{\sqrt{1 - (\lambda_0/\lambda_c)^2}}$$

Case 1 - If $\lambda_0 \ll \lambda_c$
then $\lambda_g = \lambda_0$

Case 2 - If $\lambda_0 = \lambda_c$
then $\lambda_g = \infty$

Case 3 - If $\lambda_0 > \lambda_c$
 $\lambda_g = \text{imaginary}$
(No propagation)

- TM_{m0} and TM_{0n} modes can't be physically connected they are even-scent modes.
- Ex. TM_{20} , TM_{02} ...

Wave Angle or Tilt Angle! $\rightarrow (\theta)$

Taking 2-D RWG to understand this,

where β_g = guided wave phase constant

$$= \frac{2\pi}{\lambda_g}$$

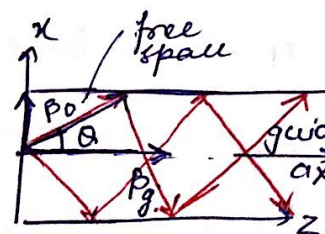
and β_0 = free space phase constant $= \frac{2\pi}{\lambda_0}$

$$\beta_g = \beta_0 \cos \theta ; \quad \frac{2\pi}{\lambda_g} = \frac{2\pi}{\lambda_0} \cos \theta$$

$$\cos \theta = \frac{\lambda_0}{\lambda_g} ; \quad \boxed{\lambda_g = \frac{\lambda_0}{\cos \theta}}$$

$$f \lambda_g = \frac{f \lambda_0}{\cos \theta}$$

(Phase velo) $v_p = \frac{c}{\cos \theta}$ — (from eq of v_p)



✓ Tilt ang
 $\cos \theta = \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$
 so $\sin \theta = \frac{f_c}{f}$
 ↑
 (from eq of v_p)

$\lambda_c - f_c$
 $\lambda_0 - f$
 $\lambda_g - f_g$

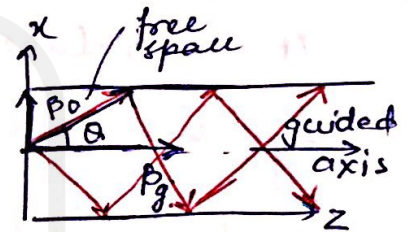
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MODES

Wave Angle or Tilt Angle! $\rightarrow (\theta)$

Taking 2-D RWG to understand this,

where β_g = guided wave phase constant
 $= \frac{2\pi}{\lambda_g}$



and β_0 = free space phase constant $= \frac{2\pi}{\lambda_0}$

$$\beta_g = \beta_0 \cos \theta \quad ; \quad \frac{2\pi}{\lambda_g} = \frac{2\pi}{\lambda_0} \cos \theta$$

$$\cos \theta = \frac{\lambda_0}{\lambda_g} \quad ; \quad \lambda_g = \frac{\lambda_0}{\cos \theta}$$

$$f \lambda_g = \frac{f \lambda_0}{\cos \theta}$$

$$(Phase \ velocity) \quad v_p = \frac{c}{\cos \theta} \quad \text{--- (from eq. of } v_p \text{)}$$

Tilt angle.

$$\cos \theta = \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

so

$$\sin \theta = \frac{f_c}{f}$$

↑

(from eq. of v_p)

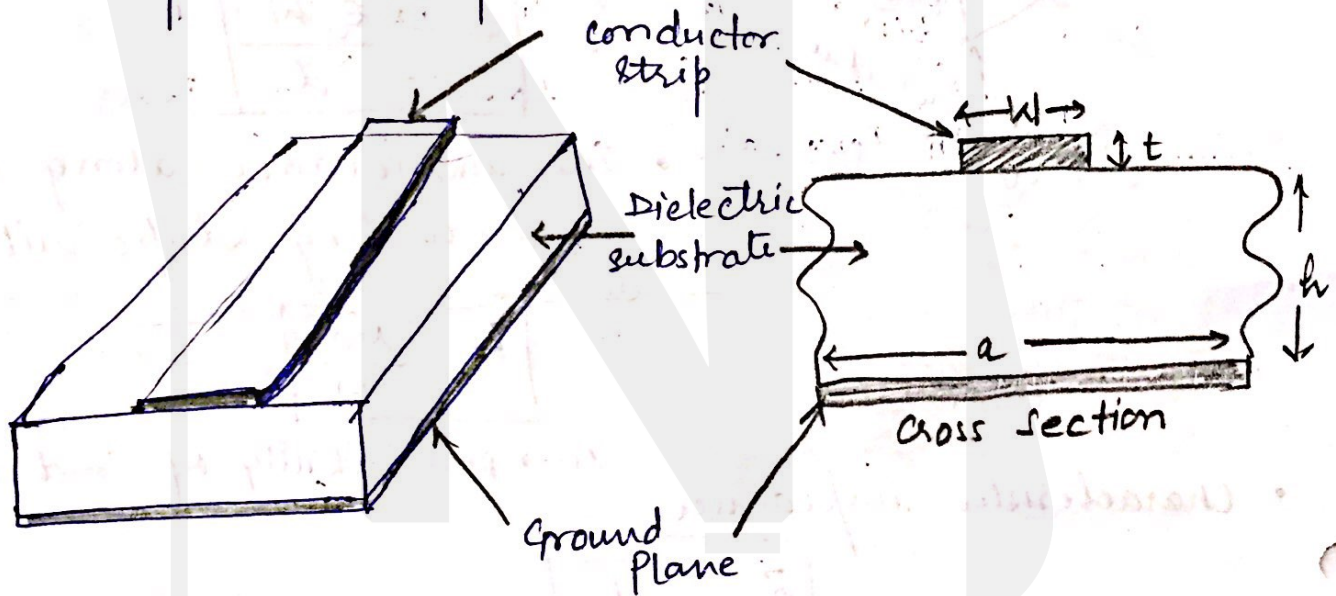
$$\lambda_c = \frac{c}{f_c}$$

$$\lambda_0 = \frac{c}{f}$$

$$\lambda_g = \frac{c}{f}$$

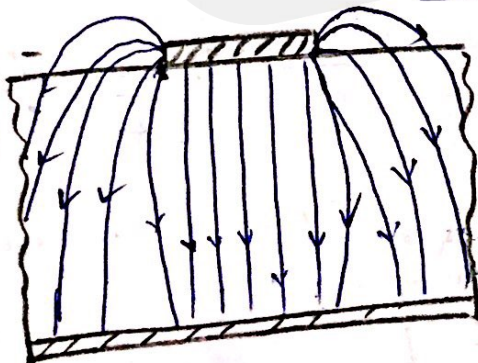
→ Micro strip TL : →

The micro strip line is also called as open strip line. Basically, it is an unsymmetrical strip line. The bottom face is metallic ground. The top face has a thin conducting strip of width ' w ' and thickness ' t '. They are separated by a low loss dielectric material. So, the upper ground plane is not present in this microstrip as compared to strip line.

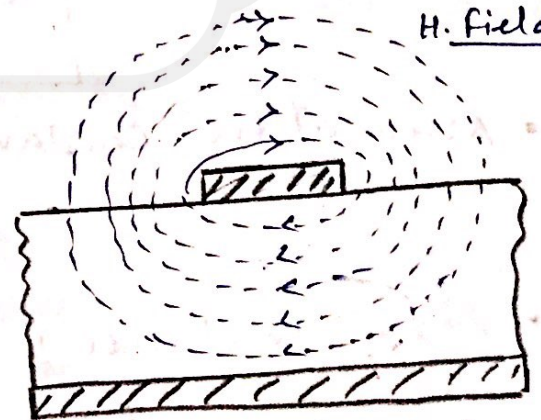


• Field Pattern:

E. field

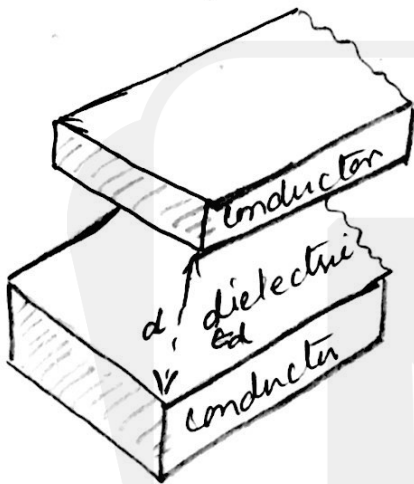


H. field



→ Types: → 1) Parallel SL
2) Coplanar SL

1) Parallel strip line: → consists of 2 perfectly parallel strips separated by a perfect dielectric slab of uniform thickness (d)
The plate width is ' W '. and relative dielectric constant of slab is ' ϵ_r '



- As 2 conducting in middle of dielectric. so it behaves like capacitor.

$$C = \frac{\epsilon_d W}{d}$$

- The inductance along the 2 conducting strips will be;

$$L = \frac{\mu d}{W}$$

μ → permeability of conductor

- Characteristic impedance;

$$Z_0 = \sqrt{\frac{L}{C}}$$

- Attenuation constant;

$$\alpha = \frac{1}{2} \left[R \sqrt{\frac{C}{L}} + G \sqrt{\frac{L}{C}} \right]$$

$(\frac{1}{R}) = \text{Transverse Conductance}$

- Phase Constant;

$$\beta = \omega \sqrt{LC}$$

2) Coplanar SL: → consists of two conducting strips on one surface with one strip grounded.

It has adv. over parallel SL b'coz its 2 strips are on the same substrate surface for convenient connections.



→ Losses in Microstrip Lines: →

- 1) Dielectric losses in substrate.
- 2) ohmic skin loss
- 3) Radiation loss.

1) Dielectric loss: → Dielectric materials have some conductivity, of course, this is very small. But due to this small σ , a dielectric attenuation α_d will be produced.

$$\alpha_d = \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}}$$

So a low loss dielectric is commonly used.

2) ohmic loss: → Due to non perfect conductors.

At MW frequency, the major attenuation factor arises due to finite conductivity of the strip conductor. ohmic losses arise due to the current on the strip.

conducting attenuation constant;

$$\alpha_c = \frac{8.68 R_s}{Z_0 W}$$

→ surface skin R.

↓

→ width of microstrip char. impedance

3) Radiation loss → due to thin open structure and any current discontinuities in strip conductor. RL depends on its geometry and dielectric constant.

The EF lines remains partially in air and partially in dielectric substrate.

Most of the EF lines are conc. in the region below the strip in the dielectric.

Therefore most of the energy is conc. in this region. MSL does not support pure TEM mode of propagation.

Advantage: → 1) Easier fabrication b'coz the entire pattern can be deposited on single dielectric substrate.

2) Better connection features

3) Both packed and unpacked IC chips can be attached easily to microstrip element due to planar nature of microstrip element.

4) Use of thin, high dielectric materials reduces the radiation loss.

5) Compact in size, light in wt, low cost, longer life.

Limitations → 1) As structure is open so may be higher radiation loss or interference due to nearby conductors.

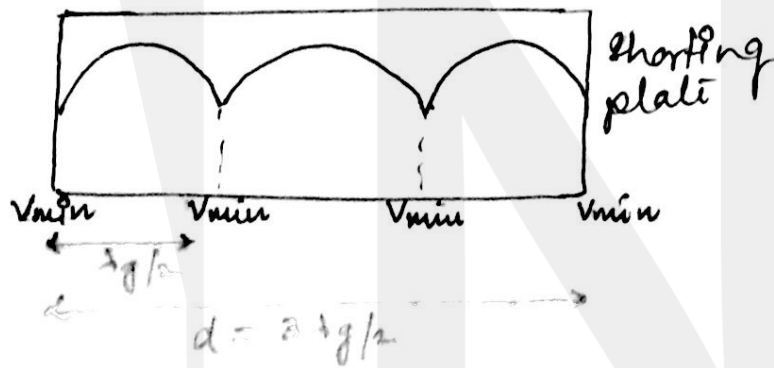
2) There is always discontinuity in EF and MF

3) Higher attenuation

4) TEM pure does not exist

5) Power handling capacity is less.

INTRODUCTION: When one end of waveguide is terminated in a shorting plate there will be reflections and hence standing wave pattern produced. When another shorting plate is kept at a distance "Multiple of λ_g " then the hollow space so formed can support a λ_g which bounce back and forth b/w the 2 shorting plates. This results in resonance and hence the hollow space is called cavity and due to resonance known as cavity Resonator.



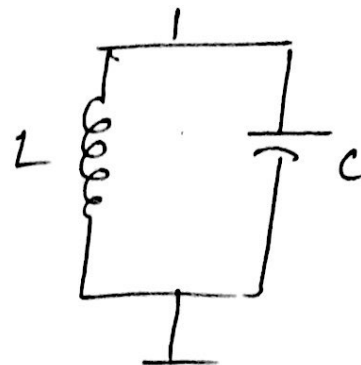
• Resonance freq can be given as:

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{c}{\epsilon}}$$

• CK can resonate at only one particular freq just as a parallel resonant CK:-

so for particular fixed $f_0 \rightarrow \lambda_0$ also fixed

$$\boxed{\lambda_0 = c/f_0}$$



$$r(k) = \frac{\text{min } V_{\text{max}}}{\text{max } V_{\text{min}}}$$

CAVITY RESONATORExpression for f_c in a Rectangular Cavity Resonator

$$k^2 = \gamma^2 + \omega^2 \mu \epsilon = A^2 + B^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

$$\omega^2 \mu \epsilon = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \gamma^2$$

for wave propagation; $\gamma = j\beta$; $\gamma^2 = -\beta^2$

$$\omega^2 \mu \epsilon = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \beta^2$$

• If a wave has to exist in cavity resonator there must be a phase change corresponding to a given wavelength; $\beta = 2\pi/\lambda_g$.

• The condition for the resonance;

$$\beta = \frac{P\pi}{d}$$

where

P - constant

d - length of cavity

• where $\beta = \frac{P\pi}{d}$ and $f = f_c$

$$\omega^2 \mu \epsilon = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{P\pi}{d}\right)^2$$

$$\left(f_c = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{P}{d}\right)^2} \right) \text{ for TE \& TM.}$$

Exp. for f_c in Circular Cavity Resonator :-

Here circular end plates are used to short both end.

a = Radius of waveguide

d = length of ~~cav~~ CWG.

condition for resonance $\beta = \frac{P\pi}{d}$



$$h^2 = \gamma^2 + \omega^2 \mu \epsilon = h_{nm}^2 = \left(\frac{P_{nm}}{a} \right)^2 \leftarrow \text{for TM}$$

$$(a h = P_{nm}) \quad = \left(\frac{P'_{nm}}{a} \right)^2 \leftarrow \text{for TE}$$

$$\omega^2 \mu \epsilon = \left(\frac{P_{nm}}{a} \right)^2 - \gamma^2$$

$$= \left(\frac{P_{nm}}{a} \right)^2 + \beta^2$$

$$= \left(\frac{P_{nm}}{a} \right)^2 + \left(\frac{p\pi}{d} \right)^2$$

dominant mode
TM₀₁₁
TE₁₀₁
TM_{nmp}
TE_{nmp}

$$f_c = \frac{1}{2\pi \sqrt{\mu \epsilon}} \left[\left(\frac{P_{nm}}{a} \right)^2 + \left(\frac{p\pi}{d} \right)^2 \right]^{1/2}$$

Free space

$$\left. \begin{aligned} f_c &= \frac{c}{2\pi} \left[\left(\frac{P_{nm}}{a} \right)^2 + \left(\frac{p\pi}{d} \right)^2 \right]^{1/2} \\ f_c &= \frac{c}{2\pi} \left[\left(\frac{P'_{nm}}{a} \right)^2 + \left(\frac{p\pi}{d} \right)^2 \right]^{1/2} \end{aligned} \right\} \begin{array}{l} \text{TM} \\ \text{TE} \end{array}$$

Application 1-

- As Tuned Ckt
- In Duplexer of Radar
- In measurement of freq.

→ Quality Factor (Q) of Cavity Resonator : →

The Q of any resonant or anti-resonant circuit is a measure of freq selectivity and defined by; $Q = \frac{\omega_0 W}{P}$

where; W = Max^m energy stored

P = Avg power loss (dissipated P)

ω_0 = Resonant freq.



$$Q = 2\pi \cdot \frac{\text{Max energy stored per cycle}}{\text{Energy dissipated for cycle}}$$

$Q = \infty \rightarrow$ (Ideal Resonator) as in perfect conductor forming the cavity then $P = 0$

- If there is more than one f_0 ; there will be different values of Q .
- Normally Q so Load coupling factor Q_L ,

$$\frac{1}{Q_L} = \frac{1}{Q_0} + \frac{1}{Q_{ext}}$$

Q_0 = unloaded cavity Quality factor.

Q_{ext} = Q due to external ohmic losses.

$$Q_0 = \omega_0 \frac{L}{R} = \frac{1}{2\pi f_0} \cdot \frac{L}{R}$$

Reflection Coefficient or

$$Q_{ext} = \frac{Q_0}{k} = \frac{\omega_0 L}{kR}$$

k = coupling Coefficient

- There are 3 type of tuned coupling in cavity Resonator :- 1) Critically Coupled 2) Under coupled 3) over Coupled.

1) CRITICALLY COUPLED: (Resonator Matched to generator); $k = 1$

$$Q_L = \frac{1}{2} Q_{ext} = \frac{1}{2} Q_0$$

2) Under Coupled: $[k < 1]$; $k = 1/p \rightarrow$ SWR

$$Q_L = \frac{p}{1+p} Q_0$$

$\rightarrow$ Terminal Voltage minimum

- 3) Over coupled cavity ($k > 1$) ($k = \beta$)
Cavity terminal are at a max^m voltage.

$$Q_L = \frac{Q_0}{1 + \beta}$$

- 4) Unloaded Q_0 :- $\frac{\text{Volume of cavity}}{\delta \times \text{surface area of cavity}}$
 $Q_0 = \frac{\text{Cross section area of cavity}}{\delta \times \text{periphery of cavity}}.$

Thus, Q of cavity can be $\uparrow$ ed by $\uparrow$ ing the size of cavity or conductivity of the walls or by $\uparrow$ ing the coupling into the cavity.

$Q \propto \text{freq} \left(f \propto \frac{1}{\text{skin depth}} \right)$

